

Hunter Ellis

(703) 953-6963 | Atlanta, GA | elliswhunter@gmail.com | github.com/hjkellis | linkedin.com/in/ellishw | ellishw.tech

Autonomy Engineer with an interest in building intelligent, capable platforms.

Experience

Autonomous Systems Engineer

Aug 2025 – Present

Georgia Tech Research Institute • Applied Systems Lab

Atlanta, GA

- Developing C++ software for collaborative autonomous UAV swarms in a component-based software engineering environment.
- Integrate, develop, and benchmark sampling-based and optimization-based motion planning algorithms over constraint manifolds leveraging The Open Motion Planning Library (OMPL).
- Implement data association algorithms for multi-object tracking (MOT) using distributed camera data from vehicles in a decentralized swarm.
- Embedded software development
- Software systems integration
- Refactored legacy code to use DDS middleware.

Robotics Research / M.S. Computer Engineering

Aug 2023 – May 2025

Virginia Tech • Control, Optimization, and Online Learning for Autonomy Lab

Blacksburg, VA

- Built a 6-DOF robotic arm and an accompanying ROS2–Gazebo simulation environment for training, evaluating, and deploying custom control algorithms.
- Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks.
- Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design.

Thrust Vector Control Intern

May 2024 – Aug 2024

Jacobs Space Exploration Group • Thrust Vector Control Test Lab

Huntsville, AL

- Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center.
- Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.

Control Research Intern

Jun 2023 – Aug 2023

G2ELab • Power Electronics Lab

Grenoble, France

- Implemented different inverter control methods for 4-leg (microgrid) topologies and tested them in Simulink and HIL.

Naval Research Enterprise Internship Program

Jun 2022 – Aug 2022

Naval Surface Warfare Center • Carderock Division

Bethesda, MD

- Developed concept hospital sea-train designs, estimated fuel consumption, and electrical power loads.

Skills

Languages: C++, Python, MATLAB/Simulink, C, Lua, $\LaTeX$

Tools and Libraries: ROS2, OMPL, DDS, Qt, Simulink, Docker, Git, FreeRTOS, Autodesk Inventor (Certified), SolidWorks, Rhino

Domain Knowledge: multi-object tracking, sampling-based motion planning, Kalman filters, factor graphs

Education

M.S. in Computer Engineering

May 2025

Virginia Tech • Bradley Department of Electrical & Computer Engineering

Blacksburg, VA

- Focus: Software and Machine Intelligence
- Research Group: Control, Optimization, and Online Learning for Autonomy Lab at UT Austin

B.S. in Electrical Engineering and Computer Engineering (Double Major)

May 2024

Virginia Tech • Bradley Department of Electrical & Computer Engineering

Blacksburg, VA

- Focus: Controls, Robotics, and Autonomy
- Design Teams: SolarCar at VT and Human Powered Submarine